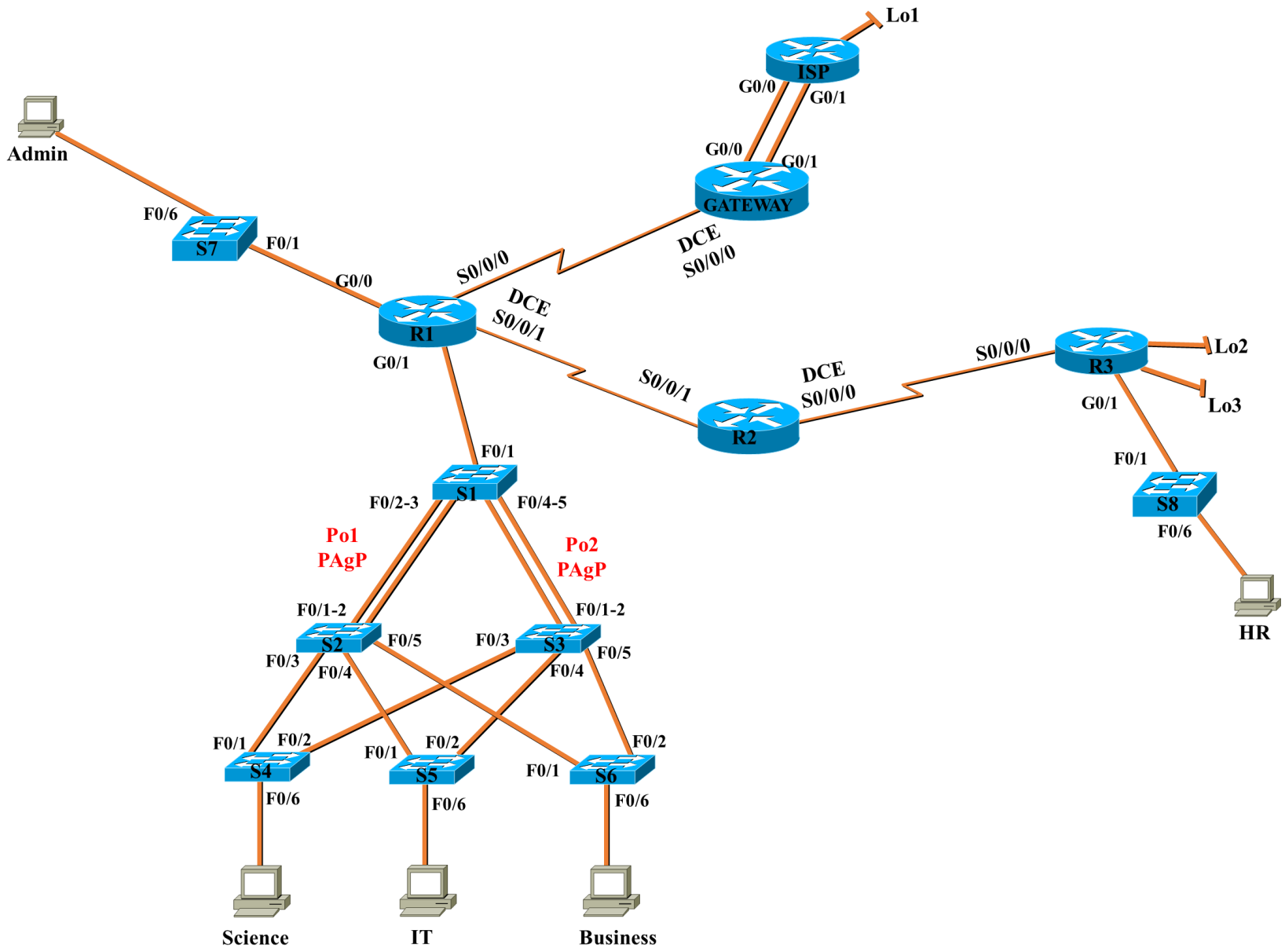
3808ICT Routing and Internetworking
Skills-based Assessment

Network Topology



Task Description

You are the network engineer for a small college. You have been asked to design and setup the network infrastructure in their building. Based on their business requirements, you have produced the following technical design:

1. A network topology diagram as shown above.

2. The network is connected to the Internet Service Provider (ISP) through its GATEWAY router. The main connection to the ISP is via the G0/0 connection, while the G0/1 connection serves as a backup. The Lo1 connection on the ISP router simulates the Internet.

3. **VLAN Table**

VLAN	Name	Devices
10	Admin	S7: F0/6
20	Science	S4: F0/6
30	IT	S5: F0/6
40	Business	S6: F0/6
50	HR	S8: F0/6
98	Management1	S7: VLAN 98
99	Management2	S1: VLAN 99 S2: VLAN 99 S3: VLAN 99 S4: VLAN 99 S5: VLAN 99 S6: VLAN 99
100	Management3	S8: VLAN 100
555	Parking_Lot	All unused ports in all switches
1000	Native	N/A

4. The network topology has several VLANs, i.e. the Admin VLAN, the Science VLAN, the IT VLAN, the Business VLAN, and the HR VLAN. There are also three management VLANs for the switched networks, i.e. Management1, Management2 and Management3 VLANs. Note that the loopback interfaces Lo2 and Lo3 are used to simulate additional LANs in the network topology. The host requirements for the LANs are shown in the table below:

LAN	Host Requirements
Admin	20 hosts
Science	130 hosts
IT	200 hosts
Business	100 hosts
HR	12 host
Lo2	40 hosts
Lo3	80 hosts
Management1	6 hosts
Management2	10 hosts
Management3	5 hosts

5. The Internet Service Provider (ISP) for the company has allocated the 132.234.24.0/22 IPv4 network address to the company. To minimise wasted address space, you will use VLSM to assign IPv4 addresses to the devices and interfaces in the network topology.

6. The default gateway IPv4 addresses in each VLAN or LAN will be assigned the first available host IPv4 address in each subnet that is created. The remaining devices' IPv4 addresses will be assigned as follows:

Device	IP Address
Admin, Science, IT, Business, HR	Will use DHCP to obtain their IPv4 addresses and other relevant network information
Lo2, Lo3	The 1st available host IPv4 address in their respective LANs
S1 VLAN 99	The 2nd available host IPv4 address in the VLAN 99 network
S2 VLAN 99	The 3rd available host IPv4 address in the VLAN 99 network
S3 VLAN 99	The 4th available host IPv4 address in the VLAN 99 network
S4 VLAN 99	The 5th available host IPv4 address in the VLAN 99 network
S5 VLAN 99	The 6th available host IPv4 address in the VLAN 99 network
S6 VLAN 99	The 7th available host IPv4 address in the VLAN 99 network
S7 VLAN 98	The 2nd available host IPv4 address in the VLAN 98 network
S8 VLAN 100	The 2nd available host IPv4 address in the VLAN 100 network

7. The IPv4 addresses for the links between each pair of routers are given below in the Addressing Table.
   Note that the "?"s (question marks) represent the IPv4 addresses that you calculated in Steps #4, #5, and #6 above.
   The IPv6 addresses for all devices are also given below in the Addressing Table.
   The Admin, Science, IT, Business, and HR end-devices will use SLAAC to obtain their IPv6 addresses and other relevant network information.
   The VLAN 98, 99, 100 networks are not assigned any IPv6 addresses.

Addressing Table

Device	Interface	IP Address/Prefix	Default Gateway
ISP	G0/0	200.100.100.1/30	N/A
		2001:DB8:ACAD::1/64	
	G0/1	200.100.100.5/30	
		2001:DB8:ACAE::1/64	
	Lo1	50.50.50.50/24	
		2001:50:50:50::50/64	
GATEWAY	G0/0	200.100.100.2/30	
		2001:DB8:ACAD::2/64	
	G0/1	200.100.100.6/30	
		2001:DB8:ACAE::2/64	
	S0/0/0	132.234.27.145/30	
		2001:DB8:CAFE:1::1/64 FE80::4	
R1	S0/0/0	132.234.27.146/30	
		2001:DB8:CAFE:1::2/64 FE80::1	
	S0/0/1	132.234.27.149/30	
		2001:DB8:CAFE:2::1/64 FE80::1	
	G0/0.10	?	
		2001:DB8:CAFE:4::1/64 FE80::1	
	G0/0.98	?	
	G0/0.1000	N/A	
	G0/1.20	?	
		2001:DB8:CAFE:5::1/64 FE80::1	
	G0/1.30	?	
		2001:DB8:CAFE:6::1/64 FE80::1	
		?	

	G0/1.40	2001:DB8:CAFE:7::1/64 FE80::1	
	G0/1.99	?	
	G0/1.1000	N/A	
R2	S0/0/0	132.234.27.153/30	
		2001:DB8:CAFE:3::1/64 FE80::2	
	S0/0/1	132.234.27.150/30	
		2001:DB8:CAFE:2::2/64 FE80::2	
R3	S0/0/0	132.234.27.154/30	
		2001:DB8:CAFE:3::2/64 FE80::3	
	G0/1.50	?	
		2001:DB8:CAFE:8::1/64 FE80::3	
	G0/1.100	?	
	G0/1.1000	N/A	
	Lo2	?	
		2001:DB8:CAFE:9::1/64	
	Lo3	?	
		2001:DB8:CAFE:A::1/64	
S1	VLAN 99	?	?
S2	VLAN 99	?	?
S3	VLAN 99	?	?
S4	VLAN 99	?	?
S5	VLAN 99	?	?
S6	VLAN 99	?	?
S7	VLAN 98	?	?
S8	VLAN 100	?	?
Admin	VLAN 10	DHCP	DHCP
		SLAAC	SLAAC
Science	VLAN 20	DHCP	DHCP
		SLAAC	SLAAC
IT	VLAN 30	DHCP	DHCP
		SLAAC	SLAAC
Business	VLAN 40	DHCP	DHCP
		SLAAC	SLAAC
HR	VLAN 50	DHCP	DHCP
		SLAAC	SLAAC

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Based on the information in the technical design above, you will implement the following Tasks:

Task 1: Perform Basic Device Configurations [15 marks]

Step 1: Set up a network that is similar to the one in the topology diagram.

Step 2: Configure the routers and switches according to the following guidelines. Use **cisco** for all passwords in this assessment, except for the enable secret password, which is **class**.

- Configure the device hostname.
- Disable DNS lookup.
- Configure an EXEC mode password.
- Encrypt the plaintext passwords.
- Create a banner that warns anyone accessing the device that unauthorised access is prohibited.
- Configure a password for console connections and enable login.
- Configure a password for vty connections and enable login.
- Configure logging synchronous for the console line and vty line.
- Configure SSH access on all routers and switches. Use username **admin** and password **cisco**. Use **cisco.com** as the domain name.

Task 2: Configure and Activate Serial and Ethernet Interface Addresses [71 marks]

Step 1: Configure the interfaces on all routers and PCs according to the IPv4 and IPv6 addresses in the Addressing Table. Assign a clock rate of 128000 to the DCE serial interfaces.

Task 3: Configure Switches [290 marks]

Step 1: Configure the relevant VLANs, access ports and trunk ports on the switches.

Step 2: Enable port security on the access ports. Configure the ports to add MAC addresses learned on the ports automatically to the running configuration. Ensure that if attempts are made to attach more than three MAC addresses to any access port, the port is shut down. In addition, assign all unused ports on the switches to the Parking_Lot VLAN and administratively deactivate them.

Step 3: Configure Management IPv4 Addresses on the switches.

Step 4: Configure the links between the switches S1 and S2, and the links between the switches S1 and S3, as EtherChannels using the PAgP protocol. On the switches, use the mode that enables the switch to actively negotiate to form a PAgP link. Use the channel group numbers as shown in the network topology diagram.

Step 5: Disable DTP negotiation on the appropriate ports on the switches.

Step 6: Configure PortFast and enable BPDU guard on the appropriate ports on the switches.

Step 7: Configure S1 as the Root Bridge for all VLANs on the switched network connected to the **R1** router.

Task 4: Configure Inter-VLAN Routing [68 marks]

Step 1: Configure inter-VLAN routing (IPv4 and IPv6) on the respective devices in the network topology.

Task 5: Configure Routing [174 marks]

Step 1: On **ISP**, add an IPv4 and IPv6 static default route with exit interface Lo1 (The Lo1 interface represents the Internet).

Step 2: Configure relevant IPv4 and IPv6 static routing between the routers **GATEWAY** and **ISP**. The **ISP** router provides Internet connectivity to all the networks behind the **GATEWAY** router. It has summarised static routes (IPv4 and IPv6) configured to reach all the networks behind the **GATEWAY** router. Use the next-hop IP address option when configuring these static routes.

Step 3: Ensure that the G0/0 link is used between the routers **GATEWAY** and **ISP**. If this link fails, the other G0/1 link should take over automatically. Use an AD of 80 for the G0/1 link.

Step 4: Configure relevant IPv4 and IPv6 static routing on the routers **GATEWAY**, **R1**, **R2**, and **R3** to ensure full network connectivity in the network topology. Use the next-hop IP address option when configuring these static routes.

Task 6: DHCP [90 marks]

Step 1: Configure DHCPv4 on the router **GATEWAY** to allocate an IPv4 address to Admin. Exclude the first five (5) usable addresses from the DHCP pool. Use **LAN10** as the pool name, and **cisco.com** as the domain name.

Step 2: Configure DHCPv4 on the router **GATEWAY** to allocate an IPv4 address to Science. Exclude the first twelve (12) usable addresses from the DHCP pool. Use **LAN20** as the pool name, and **cisco.com** as the domain name.

Step 3: Configure DHCPv4 on the router **GATEWAY** to allocate an IPv4 address to IT. Exclude the first twenty (20) usable addresses from the DHCP pool. Use **LAN30** as the pool name, and **cisco.com** as the domain name.

Step 4: Configure DHCPv4 on the router **GATEWAY** to allocate an IPv4 address to Business. Exclude the first ten (10) usable addresses from the DHCP pool. Use **LAN40** as the pool name, and **cisco.com** as the domain name.

Step 5: Configure DHCPv4 on the router **GATEWAY** to allocate an IPv4 address to HR. Exclude the first two (2) usable addresses from the DHCP pool. Use **LAN50** as the pool name, and **cisco.com** as the domain name.

Step 6: Ensure that Admin, Science, IT, Business, and HR obtain IPv6 addresses using SLAAC.

Task 7: Ping Tests [100 marks]

Admin can ping Business on IPv4

IT can ping Lo2 interface on **R3** on IPv4 and IPv6

HR can ping Lo1 interface on **ISP** on IPv4 and IPv6

Science can ping S7 on IPv4

HR can ping S4 on IPv4

IT can ping HR on IPv4

Business can ping Lo3 interface on **R3** on IPv4 and IPv6